



National Load Despatch Centre  
राष्ट्रीय भार प्रेषण केंद्र  
GRID CONTROLLER OF INDIA LIMITED  
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)  
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016  
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 20.11.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033  
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016  
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093  
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006  
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009  
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

**Sub: Daily PSP Report for the date 19.11.2024.**

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 19-नवंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 19.11.2024, is available at the NLDC website.

धन्यवाद,  
Thanks

Report for previous day Date of Reporting: 20-Nov-2024

A. Power Supply Position at All India and Regional level

|                                                                   | NR     | WR    | SR    | ER    | NER   | TOTAL  |
|-------------------------------------------------------------------|--------|-------|-------|-------|-------|--------|
| Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs) | 52391  | 61372 | 44078 | 20560 | 2662  | 181063 |
| Peak Shortage (MW)                                                | 0      | 0     | 0     | 0     | 0     | 0      |
| Energy Met (MU)                                                   | 1145   | 1495  | 1046  | 420   | 50    | 4156   |
| Hydro Gen (MU)                                                    | 114    | 38    | 84    | 26    | 17    | 278    |
| Wind Gen (MU)                                                     | 4      | 62    | 54    | -     | -     | 121    |
| Solar Gen (MU)*                                                   | 146.91 | 91.63 | 98.18 | 3.17  | 1.15  | 341    |
| Energy Shortage (MU)                                              | 0.10   | 0.00  | 0.00  | 0.00  | 0.00  | 0.10   |
| Maximum Demand Met During the Day (MW) (From NLDC SCADA)          | 57396  | 70922 | 49077 | 22005 | 2868  | 197349 |
| Time Of Maximum Demand Met                                        | 18:16  | 10:42 | 09:26 | 18:22 | 17:36 | 10:19  |

B. Frequency Profile (%)

| Region    | FVI   | < 49.7 | 49.7 - 49.8 | 49.8 - 49.9 | < 49.9 | 49.9 - 50.05 | > 50.05 |
|-----------|-------|--------|-------------|-------------|--------|--------------|---------|
| All India | 0.021 | 0.00   | 0.00        | 2.06        | 2.06   | 83.32        | 14.62   |

C. Power Supply Position in States

| Region | States               | Max.Demand Met during the day (MW) | Shortage during maximum Demand (MW) | Energy Met (MU) | Drawal Schedule (MU) | OD(+)/UD(-) (MU) | Max OD (MW) | Energy Shortage (MU) |
|--------|----------------------|------------------------------------|-------------------------------------|-----------------|----------------------|------------------|-------------|----------------------|
| NR     | Punjab               | 7442                               | 0                                   | 141.9           | 67.2                 | -0.6             | 281         | 0.00                 |
|        | Haryana              | 7247                               | 0                                   | 142.4           | 99.1                 | -0.5             | 222         | 0.00                 |
|        | Rajasthan            | 16914                              | 0                                   | 322.2           | 121.6                | -0.7             | 536         | 0.00                 |
|        | Delhi                | 3880                               | 0                                   | 71.0            | 69.3                 | -1.0             | 313         | 0.00                 |
|        | UP                   | 17882                              | 0                                   | 326.9           | 112.8                | -0.3             | 600         | 0.00                 |
|        | Uttarakhand          | 2148                               | 0                                   | 40.8            | 32.0                 | 0.6              | 113         | 0.00                 |
|        | HP                   | 1940                               | 0                                   | 35.2            | 26.7                 | -0.7             | 197         | 0.00                 |
|        | J&K(UT) & Ladakh(UT) | 2783                               | 0                                   | 56.8            | 49.5                 | 0.7              | 244         | 0.10                 |
|        | Chandigarh           | 199                                | 0                                   | 3.6             | 3.7                  | -0.1             | 58          | 0.00                 |
| WR     | Railways_NR ISTS     | 203                                | 0                                   | 4.2             | 3.9                  | 0.2              | 36          | 0.00                 |
|        | Chhattisgarh         | 4272                               | 0                                   | 87.2            | 42.5                 | 0.1              | 280         | 0.00                 |
|        | Gujarat              | 21023                              | 0                                   | 419.3           | 195.9                | -3.2             | 864         | 0.00                 |
|        | MP                   | 16970                              | 0                                   | 330.5           | 205.6                | -3.0             | 658         | 0.00                 |
|        | Maharashtra          | 27340                              | 0                                   | 582.6           | 220.1                | 5.2              | 677         | 0.00                 |
|        | Goa                  | 707                                | 0                                   | 14.5            | 13.9                 | 0.1              | 37          | 0.00                 |
|        | DNHDDPDCL            | 1330                               | 0                                   | 30.3            | 30.3                 | 0.0              | 43          | 0.00                 |
|        | AMNSIL               | 801                                | 0                                   | 18.4            | 9.1                  | 0.3              | 222         | 0.00                 |
|        | BALCO                | 529                                | 0                                   | 12.6            | 12.7                 | -0.1             | 525         | 0.00                 |
| SR     | Andhra Pradesh       | 9903                               | 0                                   | 200.3           | 47.1                 | -1.9             | 387         | 0.00                 |
|        | Telangana            | 10333                              | 0                                   | 198.9           | 110.3                | 0.1              | 857         | 0.00                 |
|        | Karnataka            | 12788                              | 0                                   | 235.4           | 71.0                 | 0.3              | 451         | 0.00                 |
|        | Kerala               | 4398                               | 0                                   | 86.7            | 68.6                 | 1.0              | 311         | 0.00                 |
|        | Tamil Nadu           | 15890                              | 0                                   | 315.1           | 181.2                | -2.0             | 572         | 0.00                 |
|        | Puducherry           | 431                                | 0                                   | 9.3             | 9.0                  | -0.2             | 24          | 0.00                 |
| ER     | Bihar                | 4453                               | 0                                   | 83.5            | 71.6                 | 1.0              | 154         | 0.00                 |
|        | DVC                  | 3165                               | 0                                   | 65.4            | -43.1                | 0.4              | 337         | 0.00                 |
|        | Jharkhand            | 1939                               | 0                                   | 36.4            | 24.3                 | -1.2             | 196         | 0.00                 |
|        | Odisha               | 4919                               | 0                                   | 91.6            | 19.0                 | -2.5             | 320         | 0.00                 |
|        | West Bengal          | 7211                               | 0                                   | 141.1           | 11.8                 | -2.6             | 155         | 0.00                 |
|        | Sikkim               | 102                                | 0                                   | 1.6             | 1.6                  | 0.0              | 21          | 0.00                 |
|        | Railways_ER ISTS     | 23                                 | 0                                   | 0.1             | 0.1                  | 0.0              | 0           | 0.00                 |
| NER    | Arunachal Pradesh    | 155                                | 0                                   | 2.8             | 2.6                  | 0.0              | 22          | 0.00                 |
|        | Assam                | 1665                               | 0                                   | 30.0            | 21.6                 | 1.5              | 131         | 0.00                 |
|        | Manipur              | 221                                | 0                                   | 2.5             | 2.5                  | 0.0              | 51          | 0.00                 |
|        | Meghalaya            | 350                                | 0                                   | 6.4             | 5.1                  | 0.0              | 258         | 0.00                 |
|        | Mizoram              | 138                                | 0                                   | 1.9             | 1.7                  | -0.6             | 30          | 0.00                 |
|        | Nagaland             | 160                                | 0                                   | 2.4             | 2.2                  | 0.0              | 36          | 0.00                 |
|        | Tripura              | 243                                | 0                                   | 4.0             | 4.9                  | 0.0              | 30          | 0.00                 |

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

|               | Bhutan | Nepal | Bangladesh | Godda -> Bangladesh |
|---------------|--------|-------|------------|---------------------|
| Actual (MU)   | 0.9    | 4.4   | -22.9      | -17.4               |
| Day Peak (MW) | 150.3  | 416.0 | -990.0     | -762.9              |

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

|              | NR    | WR     | SR    | ER     | NER | TOTAL |
|--------------|-------|--------|-------|--------|-----|-------|
| Schedule(MU) | 236.3 | -132.3 | 116.2 | -223.2 | 3.0 | 0.0   |
| Actual(MU)   | 228.5 | -123.1 | 114.8 | -226.2 | 3.5 | -2.5  |
| O/D/U/D(MU)  | -7.8  | 9.2    | -1.4  | -3.0   | 0.5 | -2.5  |

F. Generation Outage(MW)

|                | NR    | WR    | SR    | ER   | NER  | TOTAL | % Share |
|----------------|-------|-------|-------|------|------|-------|---------|
| Central Sector | 4343  | 10621 | 6008  | 3161 | 685  | 24817 | 47      |
| State Sector   | 9588  | 8809  | 7651  | 1459 | 339  | 27846 | 53      |
| Total          | 13931 | 19430 | 13659 | 4620 | 1024 | 52664 | 100     |

G. Sourcewise generation (Gross) (MU)

|                                     | NR   | WR   | SR   | ER  | NER | All India | % Share |
|-------------------------------------|------|------|------|-----|-----|-----------|---------|
| Coal                                | 651  | 1471 | 612  | 693 | 9   | 3437      | 76      |
| Lignite                             | 26   | 17   | 48   | 0   | 0   | 91        | 2       |
| Hydro                               | 114  | 38   | 84   | 26  | 17  | 278       | 6       |
| Nuclear                             | 34   | 52   | 76   | 0   | 0   | 162       | 4       |
| Gas, Naptha & Diesel                | 5    | 13   | 6    | 0   | 23  | 47        | 1       |
| RES (Wind, Solar, Biomass & Others) | 172  | 160  | 175  | 5   | 1   | 513       | 11      |
| Total                               | 1001 | 1751 | 1001 | 724 | 51  | 4529      | 100     |

|                                                                         |       |       |       |      |       |       |
|-------------------------------------------------------------------------|-------|-------|-------|------|-------|-------|
| Share of RES in total generation (%)                                    | 17.15 | 9.13  | 17.51 | 0.72 | 2.27  | 11.34 |
| Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%) | 31.92 | 14.27 | 33.44 | 4.25 | 36.40 | 21.06 |

H. All India Demand Diversity Factor

|                               |       |
|-------------------------------|-------|
| Based on Regional Max Demands | 1.024 |
| Based on State Max Demands    | 1.074 |

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

|              | Max Demand Met(MW) | Time  | Shortage(MW) |
|--------------|--------------------|-------|--------------|
| Solar hr     | 197349             | 10:19 | 0            |
| Non-Solar hr | 196606             | 18:16 | 0            |

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

\*\*Note: All generation MU figures are gross

\*\*\*Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

\*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 20-Nov-2024

| Sl No                          | Voltage Level | Line Details             | No. of Circuit | Max Import (MW) | Max Export (MW) | Import (MU) | Export (MU) | NET (MU) |
|--------------------------------|---------------|--------------------------|----------------|-----------------|-----------------|-------------|-------------|----------|
| Import/Export of ER (With NR)  |               |                          |                |                 |                 |             |             |          |
| 1                              | HVDC          | ALIPURDUAR-AGRA          | 2              | 0               | 301             | 0.0         | 7.2         | -7.2     |
| 2                              | HVDC          | PUSAULI B/B              | -              | 2               | 303             | 0.0         | 4.5         | -4.5     |
| 3                              | 765 kV        | GAYA-VARANASI            | 2              | 0               | 1153            | 0.0         | 19.1        | -19.1    |
| 4                              | 765 kV        | SASARAM-FATEHPUR         | 1              | 0               | 599             | 0.0         | 8.7         | -8.7     |
| 5                              | 765 kV        | GAYA-BALIA               | 1              | 0               | 776             | 0.0         | 14.0        | -14.0    |
| 6                              | 400 kV        | PUSAULI-VARANASI         | 1              | 39              | 181             | 0.0         | 1.8         | -1.8     |
| 7                              | 400 kV        | PUSAULI -ALLAHABAD       | 1              | 6               | 187             | 0.0         | 2.6         | -2.6     |
| 8                              | 400 kV        | MUZAFFARPUR-GORAKHPUR    | 2              | 0               | 534             | 0.0         | 8.1         | -8.1     |
| 9                              | 400 kV        | PATNA-BALIA              | 2              | 0               | 887             | 0.0         | 15.0        | -15.0    |
| 10                             | 400 kV        | NAUBATPUR-BALIA          | 2              | 0               | 353             | 0.0         | 4.7         | -4.7     |
| 11                             | 400 kV        | BIHARSHARIFF-BALIA       | 2              | 67              | 175             | 0.0         | 2.9         | -2.9     |
| 12                             | 400 kV        | MOTIHARI-GORAKHPUR       | 2              | 0               | 465             | 0.0         | 8.6         | -8.6     |
| 13                             | 400 kV        | BIHARSHARIFF-VARANASI    | 2              | 0               | 412             | 0.0         | 0.0         | 0.0      |
| 14                             | 220 kV        | SAHUPURI-KARAMNASA       | 1              | 0               | 118             | 0.0         | 0.8         | -0.8     |
| 15                             | 132 kV        | NAGAR UNTARI-RIHAND      | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| 16                             | 132 kV        | GARWAH-RIHAND            | 1              | 30              | 0               | 0.5         | 0.0         | 0.5      |
| 17                             | 132 kV        | KARMANASA-SAHUPURI       | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| 18                             | 132 kV        | KARMANASA-CHANDAULI      | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| ER-NR                          |               |                          |                |                 |                 | 0.5         | 97.9        | -97.4    |
| Import/Export of ER (With WR)  |               |                          |                |                 |                 |             |             |          |
| 1                              | 765 kV        | JHARSUGUDA-DHARAMJAIGARH | 4              | 142             | 614             | 0.0         | 7.4         | -7.4     |
| 2                              | 765 kV        | NEW RANCHI-DHARAMJAIGARH | 2              | 414             | 744             | 0.0         | 4.1         | -4.1     |
| 3                              | 765 kV        | JHARSUGUDA-DURG          | 2              | 0               | 1269            | 0.0         | 26.6        | -26.6    |
| 4                              | 400 kV        | JHARSUGUDA-RAIGARH       | 4              | 0               | 746             | 0.0         | 13.4        | -13.4    |
| 5                              | 400 kV        | RANCHI-SIPAT             | 2              | 19              | 352             | 0.0         | 3.6         | -3.6     |
| 6                              | 220 kV        | BUDHIPADAR-RAIGARH       | 1              | 0               | 116             | 0.0         | 1.7         | -1.7     |
| 7                              | 220 kV        | BUDHIPADAR-KORBA         | 2              | 6               | 155             | 0.0         | 1.5         | -1.5     |
| ER-WR                          |               |                          |                |                 |                 | 0.0         | 58.4        | -58.4    |
| Import/Export of ER (With SR)  |               |                          |                |                 |                 |             |             |          |
| 1                              | HVDC          | JEYPORE-GAZUWAKA B/B     | 2              | 0               | 206             | 0.0         | 4.9         | -4.9     |
| 2                              | HVDC          | TALCHER-KOLAR BIPOLE     | 2              | 0               | 1636            | 0.0         | 29.8        | -29.8    |
| 3                              | 765 kV        | ANGUL-SRIKAKULAM         | 2              | 0               | 2984            | 0.0         | 58.3        | -58.3    |
| 4                              | 400 kV        | TALCHER-I/C              | 2              | 657             | 0               | 13.1        | 0.0         | 13.1     |
| 5                              | 220 kV        | BALIMELA-UPPER-SILERRU   | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| ER-SR                          |               |                          |                |                 |                 | 0.0         | 92.9        | -92.9    |
| Import/Export of ER (With NER) |               |                          |                |                 |                 |             |             |          |
| 1                              | 400 kV        | BINAGURI-BONGAIGAON      | 2              | 0               | 494             | 0.0         | 7.9         | -7.9     |
| 2                              | 400 kV        | ALIPURDUAR-BONGAIGAON    | 2              | 0               | 648             | 0.0         | 9.4         | -9.4     |
| 3                              | 220 kV        | ALIPURDUAR-SALAKATI      | 2              | 0               | 99              | 0.0         | 1.1         | -1.1     |
| ER-NER                         |               |                          |                |                 |                 | 0.0         | 18.5        | -18.5    |
| Import/Export of NER (With NR) |               |                          |                |                 |                 |             |             |          |
| 1                              | HVDC          | BISWANATH CHARIALI-AGRA  | 2              | 0               | 653             | 0.0         | 15.4        | -15.4    |
| NER-NR                         |               |                          |                |                 |                 | 0.0         | 15.4        | -15.4    |
| Import/Export of WR (With NR)  |               |                          |                |                 |                 |             |             |          |
| 1                              | HVDC          | CHAMPA-KURUKSHETRA       | 2              | 0               | 1527            | 0.0         | 24.5        | -24.5    |
| 2                              | HVDC          | VINDHYACHAL B/B          | -              | 298             | 0               | 7.3         | 0.0         | 7.3      |
| 3                              | HVDC          | MUNDRA-MOHINDERGARH      | 2              | 0               | 1450            | 0.0         | 30.9        | -30.9    |
| 4                              | 765 kV        | GWALIOR-AGRA             | 2              | 0               | 1403            | 0.0         | 21.5        | -21.5    |
| 5                              | 765 kV        | GWALIOR-PHAGI            | 2              | 209             | 2095            | 0.2         | 25.7        | -25.5    |
| 6                              | 765 kV        | JABALPUR-ORAI            | 2              | 0               | 887             | 0.0         | 27.3        | -27.3    |
| 7                              | 765 kV        | GWALIOR-ORAI             | 1              | 890             | 0               | 14.7        | 0.0         | 14.7     |
| 8                              | 765 kV        | SATNA-ORAI               | 1              | 0               | 1027            | 0.0         | 20.2        | -20.2    |
| 9                              | 765 kV        | BANASKANTHA-CHITORGARH   | 2              | 1701            | 0               | 23.0        | 0.0         | 23.0     |
| 10                             | 765 kV        | VINDHYACHAL-VARANASI     | 2              | 0               | 2265            | 0.0         | 40.3        | -40.3    |
| 11                             | 400 kV        | ZERDA-KANKROLI           | 1              | 347             | 0               | 4.2         | 0.0         | 4.2      |
| 12                             | 400 kV        | ZERDA -BHINMAL           | 1              | 240             | 0               | 2.9         | 0.0         | 2.9      |
| 13                             | 400 kV        | VINDHYACHAL -RIHAND      | 1              | 965             | 0               | 22.1        | 0.0         | 22.1     |
| 14                             | 400 kV        | RAPP-SHUJALPUR           | 2              | 638             | 253             | 3.2         | 1.3         | 1.9      |
| 15                             | 400 kV        | NEEMUCH-Chittorgarh      | 2              | 697             | 254             | 3.5         | 1.5         | 2.0      |
| 16                             | 220 kV        | BHANPURA-RANPUR          | 1              | 0               | 184             | 0.0         | 3.4         | -3.4     |
| 17                             | 220 kV        | BHANPURA-MORAK           | 1              | 0               | 30              | 0.0         | 2.3         | -2.3     |
| 18                             | 220 kV        | MEHGAON-AURAIYA          | 1              | 115             | 0               | 1.6         | 0.0         | 1.6      |
| 19                             | 220 kV        | MALANPUR-AURAIYA         | 1              | 82              | 0               | 1.0         | 0.0         | 1.0      |
| 20                             | 132 kV        | GWALIOR-SAWAI MADHOPUR   | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| 21                             | 132 kV        | RAJGHAT-LALITPUR         | 2              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| WR-NR                          |               |                          |                |                 |                 | 83.8        | 198.9       | -115.2   |
| Import/Export of WR (With SR)  |               |                          |                |                 |                 |             |             |          |
| 1                              | HVDC          | BHADRAWATI B/B           | -              | 0               | 444             | 0.0         | 5.4         | -5.4     |
| 2                              | HVDC          | RAIGARH-PUGALUR          | 2              | 0               | 1004            | 0.0         | 16.6        | -16.6    |
| 3                              | 765 kV        | SOLAPUR-RAICHUR          | 2              | 987             | 947             | 4.1         | 5.7         | -1.6     |
| 4                              | 765 kV        | WARDHA-NIZAMABAD         | 2              | 0               | 1480            | 0.0         | 28.5        | -28.5    |
| 5                              | 765 kV        | WARORA-WARANGAL(NEW)     | 2              | 0               | 2245            | 0.0         | 38.0        | -38.0    |
| 6                              | 400 kV        | KOLHAPUR-KUDGI           | 2              | 1342            | 0               | 21.9        | 0.0         | 21.9     |
| 7                              | 220 kV        | KOLHAPUR-CHIKODI         | 2              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| 8                              | 220 kV        | PONDA-AMBEWADI           | 1              | 0               | 0               | 0.0         | 0.0         | 0.0      |
| 9                              | 220 kV        | XELDEM-AMBEWADI          | 1              | 0               | 108             | 2.2         | 0.0         | 2.2      |
| WR-SR                          |               |                          |                |                 |                 | 28.2        | 94.2        | -66.0    |

| INTERNATIONAL EXCHANGES |                                   |                                                                                                      |          |          | Import(+ve)/Export(-ve) |                      |
|-------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------|----------|----------|-------------------------|----------------------|
| State                   | Region                            | Line Name                                                                                            | Max (MW) | Min (MW) | Avg (MW)                | Energy Exchange (MU) |
| BHUTAN                  | ER                                | 400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)               | 200      | 80       | 115                     | 2.76                 |
|                         | ER                                | 400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW) | 155      | 58       | 111                     | 2.67                 |
|                         | ER                                | 220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)   | -275     | -177     | -216                    | -5.18                |
|                         | NER                               | 132kV GELEPHU-SALAKATI                                                                               | 20       | 0        | 8                       | 0.20                 |
|                         | NER                               | 132kV MOTANGA-RANGIA                                                                                 | 60       | -10      | 19                      | 0.45                 |
| NEPAL                   | NR                                | NEPAL IMPORT (FROM UP)                                                                               | 0        | 0        | 0                       | 0.00                 |
|                         | NR                                | 132kV MAHENDRANAGAR-TANAKPUR(NHPC)                                                                   | 0        | 0        | 0                       | 0.90                 |
|                         | ER                                | NEPAL IMPORT (FROM BIHAR)                                                                            | 0        | 0        | 0                       | 0.00                 |
|                         | ER                                | 400kV DHALKEBAR-MUZAFFARPUR 1&2                                                                      | 358      | 2        | 146                     | 3.50                 |
| BANGLADESH              | ER                                | BHERAMARA B/B HVDC (B'DESH)                                                                          | -928     | -897     | -903                    | -21.68               |
|                         | ER<br>(Isolated from Indian Grid) | 400kV GODDA_TPS-RAHANPUR (B'DESH) D/C                                                                | -763     | -552     | -726                    | -17.42               |
|                         | NER                               | 132kV COMILLA-SURAJMANI NAGAR 1&2                                                                    | -62      | 0        | -49                     | -1.18                |

Export From India (in MU)

| Country      | GNA<br>(ISGS/PPA) | T-GNA              |            |      |      |      |      |      | TOTAL |
|--------------|-------------------|--------------------|------------|------|------|------|------|------|-------|
|              |                   | BILATERAL<br>TOTAL | COLLECTIVE |      |      |      |      |      |       |
|              |                   |                    | IDAM       |      |      | RTM  |      |      |       |
|              |                   |                    | IEX        | PXIL | HPX  | IEX  | PXIL | HPX  |       |
| Bhutan       | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.18 | 0.00 | 0.00 | 0.18  |
| Nepal        | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00  |
| Bangladesh   | 22.88             | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 22.88 |
| Myanmar      | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00  |
| Total Export | 22.88             | 0.00               | 0.00       | 0.00 | 0.00 | 0.18 | 0.00 | 0.00 | 23.06 |

Import by India(in MU)

| Country      | GNA<br>(ISGA/PPA) | T-GNA              |            |      |      |      |      |      | TOTAL |
|--------------|-------------------|--------------------|------------|------|------|------|------|------|-------|
|              |                   | BILATERAL<br>TOTAL | COLLECTIVE |      |      |      |      |      |       |
|              |                   |                    | IDAM       |      |      | RTM  |      |      |       |
|              |                   |                    | IEX        | PXIL | HPX  | IEX  | PXIL | HPX  |       |
| Bhutan       | 0.92              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.92  |
| Nepal        | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 4.17 | 0.00 | 0.00 | 4.17  |
| Bangladesh   | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00  |
| Myanmar      | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00  |
| Total Import | 0.92              | 0.00               | 0.00       | 0.00 | 0.00 | 4.17 | 0.00 | 0.00 | 5.09  |

Net from India(in MU)

-ve : Export / +ve : Import

| Country    | GNA<br>(ISGS/PPA) | T-GNA              |            |      |      |       |      |      | TOTAL  |
|------------|-------------------|--------------------|------------|------|------|-------|------|------|--------|
|            |                   | BILATERAL<br>TOTAL | COLLECTIVE |      |      |       |      |      |        |
|            |                   |                    | IDAM       |      |      | RTM   |      |      |        |
|            |                   |                    | IEX        | PXIL | HPX  | IEX   | PXIL | HPX  |        |
| Bhutan     | 0.92              | 0.00               | 0.00       | 0.00 | 0.00 | -0.18 | 0.00 | 0.00 | 0.74   |
| Nepal      | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 4.17  | 0.00 | 0.00 | 4.17   |
| Bangladesh | -22.88            | 0.00               | 0.00       | 0.00 | 0.00 | 0.00  | 0.00 | 0.00 | -22.88 |
| Myanmar    | 0.00              | 0.00               | 0.00       | 0.00 | 0.00 | 0.00  | 0.00 | 0.00 | 0.00   |
| Total Net  | -21.96            | 0.00               | 0.00       | 0.00 | 0.00 | 3.99  | 0.00 | 0.00 | -17.97 |

| Date of Reporting: 20-Nov-2024                                                        |                   |                        |                     |              |               |                     |             |                     |                 |                              |                          |                                       |
|---------------------------------------------------------------------------------------|-------------------|------------------------|---------------------|--------------|---------------|---------------------|-------------|---------------------|-----------------|------------------------------|--------------------------|---------------------------------------|
| 15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA) |                   |                        |                     |              |               |                     |             |                     |                 |                              |                          |                                       |
| TIME                                                                                  | FREQUENCY<br>(Hz) | DEMAN<br>D MET<br>(MW) | NUCLEA<br>R<br>(MW) | WIND<br>(MW) | SOLAR<br>(MW) | HYDRO*<br>*<br>(MW) | GAS<br>(MW) | THERMA<br>L<br>(MW) | OTHERS*<br>(MW) | NET<br>DEMAND<br>MET<br>(MW) | TOTAL GENERATION<br>(MW) | NET TRANSNATIONAL<br>EXCHANGE<br>(MW) |
|                                                                                       |                   | (A)                    | (B)                 | (C)          | (D)           | (E)                 | (F)         | (G)                 | (H)             | (I=A-(C+D))                  | (J=B+C+D+E+F+G+H)        | (+ve) Import, (-ve) Export            |
| 0:00                                                                                  | 50.01             | 149502                 | 6309                | 6276         | 31            | 8565                | 1621        | 128465              | 1859            | 143195                       | 153126                   | -484                                  |
| 0:15                                                                                  | 49.96             | 148337                 | 6304                | 6128         | 30            | 7646                | 1639        | 128492              | 1810            | 142179                       | 152049                   | -496                                  |
| 0:30                                                                                  | 49.91             | 147553                 | 6300                | 5979         | 29            | 7647                | 1624        | 127822              | 1818            | 141545                       | 151219                   | -488                                  |
| 0:45                                                                                  | 49.91             | 146937                 | 6313                | 5856         | 28            | 7746                | 1637        | 127268              | 1808            | 141053                       | 150656                   | -474                                  |
| 1:00                                                                                  | 49.91             | 146222                 | 6298                | 5721         | 27            | 7477                | 1639        | 126963              | 1805            | 140474                       | 149930                   | -467                                  |
| 1:15                                                                                  | 49.96             | 145125                 | 6317                | 5533         | 28            | 7010                | 1619        | 126455              | 1818            | 139564                       | 148780                   | -484                                  |
| 1:30                                                                                  | 50.01             | 144267                 | 6296                | 5438         | 26            | 7015                | 1695        | 125808              | 1814            | 138803                       | 148092                   | -472                                  |
| 1:45                                                                                  | 49.96             | 144012                 | 6304                | 5456         | 27            | 7244                | 1688        | 125180              | 1853            | 138529                       | 147752                   | -460                                  |
| 2:00                                                                                  | 50.01             | 143286                 | 6311                | 5513         | 27            | 7512                | 1680        | 124109              | 1851            | 137746                       | 147003                   | -460                                  |
| 2:15                                                                                  | 49.96             | 142898                 | 6322                | 5503         | 26            | 7446                | 1686        | 123796              | 1848            | 137369                       | 146627                   | -448                                  |
| 2:30                                                                                  | 50.01             | 142482                 | 6314                | 5407         | 27            | 7439                | 1680        | 123385              | 1842            | 137048                       | 146094                   | -461                                  |
| 2:45                                                                                  | 50.01             | 141893                 | 6325                | 5375         | 26            | 7371                | 1681        | 122977              | 1877            | 136492                       | 145632                   | -476                                  |
| 3:00                                                                                  | 50.01             | 141705                 | 6317                | 5260         | 26            | 7394                | 1700        | 122986              | 1844            | 136419                       | 145527                   | -486                                  |
| 3:15                                                                                  | 50.01             | 141575                 | 6319                | 5258         | 26            | 6802                | 1758        | 123416              | 1834            | 136291                       | 145413                   | -552                                  |
| 3:30                                                                                  | 50.06             | 141407                 | 6326                | 5434         | 26            | 6812                | 1745        | 123037              | 1886            | 135947                       | 145266                   | -540                                  |
| 3:45                                                                                  | 49.96             | 141628                 | 6320                | 5609         | 26            | 6727                | 1723        | 123207              | 1834            | 135993                       | 145446                   | -500                                  |
| 4:00                                                                                  | 50.01             | 141799                 | 6312                | 5513         | 25            | 6678                | 1763        | 123576              | 1863            | 136261                       | 145730                   | -532                                  |
| 4:15                                                                                  | 49.96             | 143443                 | 6316                | 5340         | 27            | 7177                | 1751        | 124889              | 1930            | 138076                       | 147430                   | -574                                  |
| 4:30                                                                                  | 50.01             | 145051                 | 6322                | 5237         | 25            | 7510                | 1761        | 126400              | 1941            | 139789                       | 149196                   | -594                                  |
| 4:45                                                                                  | 49.96             | 146589                 | 6328                | 5134         | 27            | 7963                | 1766        | 127662              | 1896            | 141428                       | 150776                   | -634                                  |
| 5:00                                                                                  | 50.01             | 149054                 | 6312                | 5081         | 28            | 8653                | 1773        | 129614              | 1893            | 143945                       | 153354                   | -696                                  |
| 5:15                                                                                  | 49.96             | 152753                 | 6323                | 5022         | 32            | 9998                | 1757        | 132144              | 1874            | 147699                       | 157150                   | -728                                  |
| 5:30                                                                                  | 50.01             | 156015                 | 6319                | 5035         | 22            | 10664               | 1728        | 134770              | 1878            | 150958                       | 160416                   | -716                                  |
| 5:45                                                                                  | 50.06             | 159190                 | 6320                | 4966         | 38            | 11700               | 1753        | 137064              | 1907            | 154186                       | 163748                   | -792                                  |
| 6:00                                                                                  | 50.06             | 162750                 | 6311                | 4883         | 44            | 13216               | 1741        | 139177              | 1870            | 157823                       | 167242                   | -790                                  |
| 6:15                                                                                  | 50.06             | 167727                 | 6321                | 4823         | 178           | 15216               | 1745        | 141950              | 1806            | 162726                       | 172039                   | -812                                  |
| 6:30                                                                                  | 50.01             | 173262                 | 6305                | 4789         | 231           | 17765               | 1745        | 144904              | 1879            | 168242                       | 177618                   | -843                                  |
| 6:45                                                                                  | 50.01             | 178062                 | 6317                | 4974         | 525           | 19279               | 1753        | 147814              | 1817            | 172563                       | 182479                   | -865                                  |
| 7:00                                                                                  | 50.01             | 181042                 | 6320                | 4960         | 1183          | 20044               | 1752        | 149266              | 1798            | 174899                       | 185323                   | -876                                  |
| 7:15                                                                                  | 49.96             | 183964                 | 6321                | 5079         | 2322          | 20206               | 1739        | 150599              | 1815            | 176563                       | 188081                   | -868                                  |
| 7:30                                                                                  | 50.06             | 186248                 | 6317                | 5182         | 4269          | 19834               | 1757        | 150981              | 1846            | 176797                       | 190186                   | -845                                  |
| 7:45                                                                                  | 50.01             | 186830                 | 6331                | 5167         | 7152          | 16364               | 1795        | 152231              | 1888            | 174511                       | 190928                   | -863                                  |
| 8:00                                                                                  | 50.06             | 186514                 | 6419                | 4990         | 10667         | 14702               | 1792        | 150055              | 1964            | 170857                       | 190589                   | -875                                  |
| 8:15                                                                                  | 50.06             | 187976                 | 6500                | 4975         | 14892         | 13321               | 1789        | 148417              | 1995            | 168109                       | 191889                   | -857                                  |
| 8:30                                                                                  | 50.06             | 188294                 | 6481                | 4914         | 18932         | 11773               | 1757        | 146214              | 2068            | 164448                       | 192139                   | -821                                  |
| 8:45                                                                                  | 50.01             | 190402                 | 6461                | 4841         | 22762         | 10948               | 1726        | 145520              | 2115            | 162799                       | 194373                   | -798                                  |
| 9:00                                                                                  | 50.06             | 191643                 | 6462                | 4804         | 26661         | 10459               | 1731        | 143466              | 2068            | 160178                       | 195651                   | -787                                  |
| 9:15                                                                                  | 50.01             | 194735                 | 6469                | 4433         | 30149         | 10113               | 1727        | 143559              | 2085            | 160153                       | 198535                   | -785                                  |
| 9:30                                                                                  | 50.06             | 195793                 | 6466                | 4305         | 33196         | 9825                | 1744        | 142050              | 2087            | 158292                       | 199673                   | -774                                  |
| 9:45                                                                                  | 50.01             | 196783                 | 6472                | 4337         | 36087         | 9045                | 1729        | 140530              | 2089            | 156359                       | 200289                   | -807                                  |
| 10:00                                                                                 | 50.06             | 196827                 | 6460                | 4380         | 38389         | 9034                | 1723        | 138166              | 2088            | 154058                       | 200240                   | -804                                  |
| 10:15                                                                                 | 50.01             | 196800                 | 6460                | 4456         | 40648         | 8389                | 1724        | 136386              | 2174            | 151696                       | 200237                   | -789                                  |
| 10:30                                                                                 | 50.01             | 196717                 | 6468                | 4516         | 42556         | 7764                | 1731        | 134855              | 2080            | 149645                       | 199970                   | -753                                  |
| 10:45                                                                                 | 50.01             | 196632                 | 6464                | 4330         | 44438         | 7768                | 1734        | 133042              | 2100            | 147864                       | 199876                   | -756                                  |
| 11:00                                                                                 | 50.06             | 196072                 | 6443                | 4215         | 45584         | 7598                | 1733        | 131651              | 2167            | 146273                       | 199391                   | -747                                  |
| 11:15                                                                                 | 50.01             | 194009                 | 6452                | 3954         | 46607         | 7127                | 1728        | 129751              | 2104            | 143448                       | 197723                   | -745                                  |
| 11:30                                                                                 | 49.96             | 194295                 | 6433                | 3642         | 47470         | 7203                | 1716        | 129133              | 2141            | 143183                       | 197738                   | -745                                  |
| 11:45                                                                                 | 49.96             | 193798                 | 6438                | 3187         | 48323         | 7256                | 1725        | 128427              | 2123            | 142288                       | 197479                   | -763                                  |
| 12:00                                                                                 | 50.01             | 193337                 | 6427                | 2984         | 48987         | 7241                | 1715        | 127604              | 2196            | 141366                       | 197154                   | -709                                  |
| 12:15                                                                                 | 49.95             | 192071                 | 6443                | 2765         | 48848         | 6981                | 1711        | 126202              | 2120            | 140458                       | 195070                   | -698                                  |
| 12:30                                                                                 | 49.95             | 190643                 | 6428                | 2575         | 49171         | 6730                | 1716        | 124942              | 2127            | 138897                       | 193689                   | -708                                  |
| 12:45                                                                                 | 49.95             | 189134                 | 6442                | 2489         | 49161         | 6514                | 1715        | 124334              | 2119            | 137484                       | 192774                   | -680                                  |
| 13:00                                                                                 | 50.00             | 187429                 | 6443                | 2477         | 48806         | 6379                | 1713        | 123724              | 2113            | 136146                       | 191655                   | -717                                  |
| 13:15                                                                                 | 50.00             | 183972                 | 6433                | 2835         | 48023         | 5948                | 1697        | 120382              | 2083            | 133114                       | 187401                   | -767                                  |
| 13:30                                                                                 | 50.00             | 183419                 | 6428                | 2975         | 47028         | 5976                | 1696        | 120856              | 2101            | 133416                       | 187060                   | -742                                  |
| 13:45                                                                                 | 50.03             | 184276                 | 6424                | 3051         | 45919         | 6188                | 1727        | 122288              | 2098            | 135306                       | 187695                   | -791                                  |
| 14:00                                                                                 | 50.03             | 184655                 | 6428                | 3022         | 45027         | 6167                | 1733        | 123383              | 2091            | 136606                       | 187851                   | -808                                  |
| 14:15                                                                                 | 49.98             | 184266                 | 6422                | 3198         | 43217         | 6196                | 1738        | 124893              | 2102            | 137851                       | 187766                   | -816                                  |
| 14:30                                                                                 | 49.99             | 186526                 | 6427                | 3418         | 41339         | 6695                | 1740        | 128313              | 2131            | 141769                       | 190063                   | -867                                  |
| 14:45                                                                                 | 50.04             | 187022                 | 6430                | 3503         | 38974         | 7094                | 1738        | 130730              | 2029            | 144545                       | 190498                   | -826                                  |
| 15:00                                                                                 | 50.09             | 187421                 | 6432                | 3620         | 36502         | 7412                | 1741        | 133152              | 2004            | 147299                       | 190863                   | -875                                  |
| 15:15                                                                                 | 50.03             | 186704                 | 6428                | 3654         | 33399         | 7608                | 1739        | 135539              | 1994            | 149651                       | 190361                   | -858                                  |
| 15:30                                                                                 | 50.03             | 187454                 | 6432                | 3830         | 30108         | 8366                | 1734        | 138571              | 2023            | 153516                       | 191064                   | -843                                  |
| 15:45                                                                                 | 49.98             | 187607                 | 6421                | 3764         | 26681         | 8886                | 1843        | 141440              | 1956            | 157162                       | 190991                   | -865                                  |
| 16:00                                                                                 | 50.03             | 187940                 | 6415                | 3759         | 22988         | 10332               | 1989        | 143868              | 1947            | 161193                       | 191298                   | -882                                  |
| 16:15                                                                                 | 49.93             | 186440                 | 6422                | 3639         | 18971         | 10776               | 2526        | 145625              | 1943            | 163830                       | 189902                   | -879                                  |
| 16:30                                                                                 | 49.98             | 187442                 | 6420                | 3496         | 14795         | 12525               | 2808        | 149387              | 1912            | 169151                       | 191343                   | -886                                  |
| 16:45                                                                                 | 49.98             | 188779                 | 6418                | 3375         | 10702         | 14860               | 3076        | 152418              | 1936            | 174702                       | 192785                   | -812                                  |
| 17:00                                                                                 | 49.98             | 190045                 | 6419                | 3301         | 7101          | 17027               | 3161        | 155444              | 1874            | 179643                       | 194327                   | -867                                  |
| 17:15                                                                                 | 50.03             | 189636                 | 6424                | 3264         | 4299          | 18622               | 2823        | 156700              | 1850            | 182073                       | 193982                   | -894                                  |
| 17:30                                                                                 | 50.03             | 192091                 | 6433                | 3391         | 2353          | 21005               | 3262        | 158199              | 1866            | 186347                       | 196509                   | -931                                  |
| 17:45                                                                                 | 49.98             | 194546                 | 6429                | 3568         | 1237          | 23309               | 3380        | 159001              | 1864            | 189741                       | 198788                   | -917                                  |
| 18:00                                                                                 | 50.03             | 195448                 | 6423                | 3717         | 977           | 24459               | 3404        | 158802              | 1875            | 190754                       | 199657                   | -895                                  |
| 18:15                                                                                 | 49.98             | 196371                 | 6427                | 3968         | 785           | 25337               | 3395        | 158815              | 1820            | 191618                       | 200547                   | -888                                  |
| 18:30                                                                                 | 50.03             | 195544                 | 6459                | 4262         | 779           | 25376               | 2078        | 158970              | 1851            | 190503                       | 199775                   | -863                                  |
| 18:45                                                                                 | 49.98             | 193316                 | 6423                | 4466         | 700           | 24061               | 1814        | 158246              | 1817            | 188150                       | 197527                   | -801                                  |
| 19:00                                                                                 | 50.03             | 190056                 | 6437                | 4616         | 660           | 23492               | 1824        | 155296              | 1869            | 184780                       | 194194                   | -811                                  |
| 19:15                                                                                 | 50.03             | 186590                 | 6435                | 4921         | 658           | 21538               | 1622        | 153601              | 1835            | 181011                       | 190610                   | -741                                  |
| 19:30                                                                                 | 50.08             | 183760                 | 6439                | 5047         | 662           | 19918               | 1632        | 151964              | 1825            | 178051                       | 187487                   | -751                                  |
| 19:45                                                                                 | 50.08             | 179878                 | 6441                | 5178         | 660           | 18099               | 1630        | 149851              | 1854            | 174040                       | 183713                   | -723                                  |
| 20:00                                                                                 | 50.08             | 175593                 | 6451                | 5348         | 657           | 15984               | 1789        | 147508              | 1823            | 169588                       | 179560                   | -707                                  |
| 20:15                                                                                 | 50.03             | 172023                 | 6443                | 5558         | 57            | 14657               | 1639        | 145116              | 1916            | 166408                       | 175386                   | -708                                  |
| 20:30                                                                                 | 50.03             | 169526                 | 6433                | 5612         | 56            | 13808               | 1618        | 143404              | 1936            | 163858                       | 172867                   | -693                                  |
| 20:45                                                                                 | 50.03             | 167440                 | 6443                | 5675         | 55            | 13236               | 1634        | 141682              | 1936            | 161710                       | 170661                   | -683                                  |
| 21:00                                                                                 | 49.98             | 165139                 | 6446                | 5768         | 16            | 12917               | 1626        | 139874              | 1960            | 159355                       | 168607                   | -649                                  |
| 21:15                                                                                 | 50.03             | 162440                 | 6444                | 5761         | 13            | 12413               | 1626        | 137513              | 1995            | 156666                       | 165765                   | -598                                  |
| 21:30                                                                                 | 50.03             | 159948                 | 6431                | 5777         | 15            | 12160               | 1630        | 135762              | 1929            | 154156                       | 163704                   | -560                                  |
| 21:45                                                                                 | 49.98             | 158909                 | 6438                | 5886         | 15            | 11218               | 1635        | 134745              | 1976            | 153008                       | 161913                   | -542                                  |
| 22:00                                                                                 | 50.08             | 156386                 | 6425                | 6126         | 16            | 9502                | 1624        | 133896              | 1953            | 150244                       | 159542                   | -536                                  |
| 22:15                                                                                 | 49.98             | 156792                 | 6426                | 6220         | 19            | 9241                | 1636        | 134469              | 1919            | 150553                       | 159930                   | -581                                  |
| 22:30                                                                                 | 49.98             | 155950                 | 6435                | 6357         | 17            | 9353                | 1631        | 133534              | 1943            | 149576                       | 159270                   | -522                                  |
| 22:45                                                                                 |                   |                        |                     |              |               |                     |             |                     |                 |                              |                          |                                       |